Developing Mutation Detection Algorithms for SARS-CoV-2 Covid 19 Genomic Sequences

PROJECT REPORT

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BONAFIDE CERTIFICATE

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DECLARATION

I, the undersigned, hereby declare that this thesis entitled **"Developing Mutation Detection Algorithms for SARS-CoV-2 Covid 19 Genomic Sequences"**, is the record of the original work done by Abhishek Karthik J (CB.SC.U4AIE23010), Keerthivasan S V (CB.SC.U4AIE23037), Mothishwaran M P (CB.SC.U4AIE23041), and Mopuru Sai Bavesh Reddy (CB.SC.U4AIE23044) under the guidance of **Guide Name 1**, Designation and **Guide Name 2**, Designation, Amrita School of Artificial Intelligence, Coimbatore. To the best of my knowledge this work has not formed the basis for the award of any degree/diploma/ associateship/fellowship/or a similar award to any candidate in any University.

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Contents

Acknowledgement **iv**

List of Figures **v**

List of Tables **vi**

List of Abbreviations **vii**

Abstract **viii**

1 Introduction **1**

1.1 Literature Survey 2

1.2 Problem statement 3

1.3 Objectives 4

2 Background **6**

2.1 SARS-CoV-2 Genomic Characteristics 6

2.2 Multiple Sequence Alignment 6

2.2.1 MAFFT Algorithm 7

2.2.2 PRANK Algorithm 9

2.3 Phylogenetic Analysis 11

2.4 Needleman-Wunsch Algorithm 12

3 Proposed Work **14**

3.1 Data Collection and Preprocessing 14

3.1.1 Data Collection Process 14

3.1.2 Sequence Preprocessing 15

3.2 Implementation of MAFFT Algorithm 16

3.2.1 Numerical Encoding and FFT Application 16

3.2.2 Distance Matrix Computation 17

3.2.3 Guide Tree Construction 18

3.2.4 Progressive Alignment 18

3.2.5 Iterative Refinement 19

3.3 Implementation of PRANK Algorithm 20

3.3.1 Pairwise Distance Calculation 20

3.3.2 UPGMA Tree Construction 20

3.3.3 Pairwise Alignment with Indel Distinction 21

3.3.4 Progressive Phylogeny-Aware Alignment 22

3.4 Global Alignment Implementation 22

3.4.1 Initialization 23

3.4.2 Matrix Filling 23

3.4.3 Traceback 24

3.4.4 Identity Calculation 24

3.5 Phylogenetic Analysis 25

3.5.1 Multiple Sequence Alignment Processing 25

3.5.2 Distance Calculation 25

3.5.3 Tree Construction 26

3.5.4 Tree Visualization and Analysis 26

3.6 Consensus Sequence Comparison 27

3.6.1 Consensus Sequence Generation 27

3.6.2 Fast Identity Comparison 28

3.6.3 Identity Matrix Construction 28

4 Results and Discussion **30**

4.1 Sequence Alignment Results 30

4.2 Consensus Sequence Comparison 31

4.3 Phylogenetic Analysis 32

4.4 Regional Variation and Public Health Implications 33

5 Conclusion **35**

References **38**

List of Publications based on this research work **40**

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List of Figures

List of Tables

List of Abbreviations

SARS-CoV-2	-	Severe Acute Respiratory Syndrome Coronavirus 2
MSA	-	Multiple Sequence Alignment
MAFFT	-	Multiple Alignment using Fast Fourier Transform
PRANK	-	Probabilistic Alignment Kit
FFT	-	Fast Fourier Transform
CSV	-	Comma-Separated Values
UPGMA	-	Unweighted Pair Group Method with Arithmetic Mean
NJ	-	Neighbor-Joining
HRM	-	High-Resolution Melting
RT-PCR	-	Reverse Transcription Polymerase Chain Reaction

Abstract

This research project focuses on developing and implementing multiple sequence alignment (MSA) algorithms for detecting mutations in SARS-CoV-2 genomic sequences. The emergence of SARS-CoV-2 variants with potentially increased transmissibility and impact on vaccine efficacy highlights the need for robust computational methods to analyze genomic variations.

Our study implements two MSA algorithms from scratch: Multiple Alignment using Fast Fourier Transform (MAFFT) and Probabilistic Alignment Kit (PRANK). We apply these algorithms to analyze genomic sequences collected from three U.S. states: California, Texas, and New York. The implementation includes data collection, preprocessing, alignment, consensus sequence generation, and phylogenetic analysis.

The research demonstrates significant differences in mutation patterns across geographic regions. Our analysis reveals that California and Texas sequences exhibit high similarity (90.08%) using MAFFT, while New York sequences show greater divergence. Contrastingly, PRANK-based analysis indicates more uniform divergence across all three states. The study also analyzes evolutionary drift from the Wuhan reference strain through phylogenetic tree construction.

This work contributes to our understanding of SARS-CoV-2 genomic variations and demonstrates the efficacy of computational approaches in tracking viral evolution. The methodologies developed can be extended to other viral genomic analyses and may aid in epidemiological research and public health response strategies.

Chapter 1

Introduction

The emergence of SARS-CoV-2, the virus responsible for the COVID-19 pandemic, has prompted extensive research efforts to understand its genomic structure, evolution, and mutations. As viruses replicate, they accumulate genetic changes that can affect their transmissibility, virulence, and response to vaccines or treatments. Tracking these mutations is crucial for public health surveillance and pandemic response strategies.

This research focuses on developing and implementing multiple sequence alignment (MSA) algorithms to detect and analyze mutations in SARS-CoV-2 genomic sequences. We specifically examine sequences from three U.S. states—California, Texas, and New York—to understand regional variations in viral evolution.

The study implements two MSA algorithms from scratch: Multiple Alignment using Fast Fourier Transform (MAFFT) and Probabilistic Alignment Kit (PRANK). These algorithms allow for efficient and accurate alignment of multiple genomic sequences, enabling the identification of conserved regions and mutations.

Our methodology encompasses several key stages: data collection and filtering, sequence preprocessing and cleaning, multiple sequence alignment using MAFFT and

PRANK, consensus sequence generation, and phylogenetic analysis. By comparing the results from different algorithms and across different regions, we aim to gain insights into the evolutionary patterns of SARS-CoV-2.

The findings of this research contribute to our understanding of SARS-CoV-2 genomic variations and demonstrate the effectiveness of computational approaches in tracking viral evolution. The methodologies developed can be extended to other viral genomic analyses and may aid in epidemiological research and public health response strategies.

1.1 Literature Survey

Our review of recent literature reveals significant advancements in SARS-CoV-2 mutation detection and analysis:

- "An Update on Detection Technologies for SARS-CoV-2 Variants of Concern" highlighted the effectiveness of RT-PCR assays in identifying variants by targeting known mutation sites, enabling quicker detection compared to sequencing methods.

- "Enhanced Detection and Molecular Modeling of Adaptive Mutations in SARS-CoV-2" (2023) applied a site-by-site analysis to identify nucleotide and amino acid sites under selection, providing insights into adaptive mutations that could influence viral behavior.

- "Global Landscape of SARS-CoV-2 Mutations and Conserved Regions" (2023)

conducted a comprehensive analysis of SARS-CoV-2 genomes to identify mutations and conserved regions, providing a systematic resource for novel sequence features, aiding in the development of vaccines and therapeutics by highlighting conserved regions less prone to mutation.

- "Tracking Mutational Semantics of SARS-CoV-2 Genomes" (2022) employed natural language processing algorithms to analyze SARS-CoV-2 genome mutations, demonstrating that NLP-based approaches can effectively process genomic data, revealing characteristics and evolutionary patterns of the virus.

- "Rapid Detection of Predominant SARS-CoV-2 Variants Using High-Resolution Melting Analysis" (2023) developed a multiplex high-resolution melting (HRM) method for SARS-CoV-2 detection and mutation site identification, establishing a one-step multiplex HRM assay capable of detecting SARS-CoV-2 and identifying specific mutation sites, facilitating rapid variant analysis.

Our research builds upon these foundations while implementing novel approaches to multiple sequence alignment and phylogenetic analysis of SARS-CoV-2 genomes.

1.2 Problem statement

The rapid evolution of SARS-CoV-2 has resulted in multiple variants with potentially different transmissibility, virulence, and response to vaccines. Traditional methods for genomic sequence analysis often rely on existing tools without exploring the underlying algorithms or customizing them for SARS-CoV-2-specific characteristics. This research

addresses the need for:

- Implementation of multiple sequence alignment algorithms from scratch to provide greater transparency and customization for SARS-CoV-2 analysis
- Comparison of different alignment approaches (MAFFT and PRANK) to determine their relative effectiveness for viral mutation detection
- Regional analysis of SARS-CoV-2 sequences to understand geographic patterns of viral evolution
- Development of computational methods to identify mutations and assess their potential impact

1.3 Objectives

- Develop MSA-based approaches (MAFFT and PRANK) from scratch to analyze SARS-CoV-2 genomic sequences
- Generate consensus viral sequences from multiple samples across California, New York, and Texas
- Compare these sequences with the Wuhan reference strain to track evolutionary divergence
- Identify mutation hotspots and regional variation patterns
- Construct phylogenetic trees to visualize evolutionary relationships between regional strains

- Implement efficient algorithms for sequence alignment, mutation detection, and phylogenetic analysis

The chapter is organized as follows: Section 2 describes the background information of the network architecture Section 3 describes the proposed work, and the thesis is concluded in section 4.

Chapter 2

Background

2.1 SARS-CoV-2 Genomic Characteristics

SARS-CoV-2 is a positive-sense single-stranded RNA virus with a genome of approximately 30,000 nucleotides. The genome encodes structural proteins including spike (S), envelope (E), membrane (M), and nucleocapsid (N) proteins, as well as several non-structural proteins with functions in viral replication and host immune evasion.

The virus exhibits a mutation rate typical of RNA viruses, which contributes to its genetic diversity and evolution. These mutations can affect viral properties such as transmissibility, virulence, and immune escape, making genomic surveillance an essential component of pandemic response.

2.2 Multiple Sequence Alignment

Multiple Sequence Alignment (MSA) is a computational method used to align three or more biological sequences (DNA, RNA, or protein) to identify regions of similarity. These alignments can reveal conserved domains, functional regions, and evolutionary relationships between sequences.

2.2.1 MAFFT Algorithm

MAFFT (Multiple Alignment using Fast Fourier Transform) is a widely-used algorithm for multiple sequence alignment. It aligns nucleotide or protein sequences to identify regions of similarity that may indicate functional, structural, or evolutionary relationships between the sequences.

Key features of MAFFT include:

- Speed: MAFFT is faster than many traditional alignment methods due to the use of FFT.
- Accuracy: It provides high-quality alignments by considering both local and global sequence similarities.
- Versatility: MAFFT can handle a wide range of sequence lengths and numbers of sequences.

The MAFFT algorithm consists of four main steps:

1. **Compute the Similarity Matrix using FFT (O(n log n))**: Instead of directly comparing sequences character by character, MAFFT uses Fast Fourier Transform to convert sequences into frequency representations. This is done by first encoding nucleotides numerically (e.g., A=1, C=2, G=3, T=4), then applying FFT to transform these numerical sequences into frequency components. The similarity between sequences can then be calculated by comparing these fre-

quency spectra, which is much faster than direct comparison, especially for long sequences.

2. **Construct the Guide Tree ($O(N^2)$)**: Using the similarity matrix obtained in the previous step, MAFFT builds a hierarchical clustering of sequences called a guide tree. This tree determines the order in which sequences will be aligned. Sequences with higher similarity are clustered first, creating a strategy for progressive alignment. This is typically implemented using algorithms like UPGMA (Unweighted Pair Group Method with Arithmetic Mean) or Neighbor-Joining.

3. **Progressive Alignment ($O(N^2)$)**: Sequences are aligned progressively following the order defined by the guide tree. Starting from the most similar sequences, pairwise alignments are performed and extended as we move up the tree. This approach ensures that the most similar sequences are aligned first, reducing the chances of misalignment.

4. **Iterative Refinement ($O(N^3)$)**: To improve the initial alignment, MAFFT employs an iterative refinement strategy. This involves removing one sequence at a time from the current alignment, realigning it to the remaining profile, and checking if the overall alignment score improves. This process is repeated until no further improvement is observed or a maximum number of iterations is reached.

2.2.2 PRANK Algorithm

PRANK (Probabilistic Alignment Kit) is a phylogeny-aware multiple sequence alignment algorithm. Unlike traditional methods, PRANK distinguishes between insertions and deletions during the alignment process, which helps in more accurately representing the evolutionary history of sequences.

Key features of PRANK include:

- Phylogeny-aware: PRANK uses evolutionary information to guide the alignment process
- Accurate indel handling: It distinguishes between insertions and deletions, reducing alignment errors
- Evolutionary modeling: Incorporates evolutionary models to better represent sequence changes over time
- Reduced alignment bias: Avoids systematic errors common in progressive alignment methods

The PRANK algorithm consists of four main steps:

1. **Construct a Guide Tree ($\mathbf{O(N^2)}$)**: PRANK begins by estimating evolutionary distances between sequences, typically using metrics like Hamming distance or more sophisticated evolutionary models. These distances are then used to build a phylogenetic guide tree that represents the evolutionary relationships between

sequences. The guide tree is crucial as it determines not just the order of alignment but also helps in tracking evolutionary events.

2. **Distinguish Insertions and Deletions ($O(N^2)$)**: A key innovation in PRANK is its ability to differentiate between insertions and deletions. As the alignment progresses, PRANK tracks the history of indels (insertions and deletions) and labels new gaps as either insertions or deletions based on the guide tree information. This prevents the same gap from being penalized multiple times in downstream alignments, which is a common problem in traditional progressive alignment methods.

3. **Phylogeny-Aware Progressive Alignment ($O(N^3)$)**: Sequences are aligned following the topology of the guide tree. At each internal node of the tree, PRANK reconstructs ancestral sequences based on the descendants. When aligning sequences or profiles, the algorithm considers the evolutionary changes that might have occurred since their divergence from their common ancestor, leading to more biologically accurate alignments.

4. **Using Posterior Probabilities for Alignment Decisions ($O(N^2)$)**: PRANK uses a probabilistic framework to make alignment decisions. It calculates the probability of different evolutionary scenarios (e.g., insertion, deletion, substitution) and chooses the one with the highest posterior probability. This approach minimizes systematic errors in gap placement, particularly in regions with multiple indels, resulting in alignments that better reflect the true evolutionary history.

2.3 Phylogenetic Analysis

Phylogenetic analysis is a method used to study the evolutionary relationships between different species, organisms, or genes. In the context of SARS-CoV-2, phylogenetic analysis helps in tracking the spread and evolution of the virus over time and across geographical regions.

Key methods in phylogenetic analysis include:

- Distance-based methods (e.g., Neighbor-Joining): These methods calculate a distance matrix between sequences and use it to construct a tree. The Neighbor-Joining algorithm, in particular, starts with a star-like tree and iteratively joins the closest pairs of nodes, recalculating distances at each step.

- Maximum likelihood: This method evaluates the probability of observing the given sequences under a specific evolutionary model. It selects the tree topology and branch lengths that maximize this probability.

- Bayesian inference: This approach uses Bayesian statistics to estimate the posterior probability of a tree given the sequence data and prior information about evolutionary processes.

Phylogenetic trees provide a visual representation of the evolutionary relationships between sequences, with branch lengths often indicating the amount of genetic change or time since divergence.

2.4 Needleman-Wunsch Algorithm

The Needleman-Wunsch algorithm is a dynamic programming algorithm used for global sequence alignment. It identifies the optimal alignment of two sequences over their entire length by maximizing the similarity score.

The algorithm consists of three main steps:

1. **Initialization**: Create a scoring matrix with dimensions (m+1) × (n+1), where m and n are the lengths of the two sequences. Initialize the first row and column with gap penalties, typically using a linear gap model where each gap receives a constant penalty.

2. **Matrix Filling**: Fill the matrix cell by cell, starting from the top-left corner. For each cell (i,j), calculate the score as the maximum of three possibilities:

   - Diagonal move (match/mismatch): Score from cell (i-1, j-1) plus the score for aligning the characters at positions i and j (positive for match, negative for mismatch)
   - Horizontal move (gap in first sequence): Score from cell (i, j-1) plus the gap penalty
   - Vertical move (gap in second sequence): Score from cell (i-1, j) plus the gap penalty

3. **Traceback**: Starting from the bottom-right cell of the matrix, trace back through the matrix to reconstruct the optimal alignment. At each step, move to the

cell that contributed to the current cell's score (diagonal for match/mismatch, horizontal or vertical for gaps). This process continues until reaching the top-left cell, yielding the complete alignment.

This algorithm is particularly useful for aligning sequences of similar length and structure, such as SARS-CoV-2 genomes from different samples or regions.

Chapter 3

Proposed Work

3.1 Data Collection and Preprocessing

Our dataset contains genomic sequences of SARS-CoV-2, with detailed metadata for mutation detection and strain classification. Key features include:

- **Genomic Metadata**: Accession ID, BioProject & BioSample, Completeness, Collection Date & Location, Submitter Information
- **Genomic Sequences**: Sequence ID, Nucleotide Sequence
- **Variant & Strain Classification**: Virus Name & TaxID, Pangolin Classification, Mutations (SNPs, insertions, deletions)

3.1.1 Data Collection Process

Our data collection process focused on extracting SARS-CoV-2 genomic sequences from three target U.S. states: California, Texas, and New York. We implemented a robust CSV parsing system that could:

1. **Automatically detect CSV delimiters**: The system analyzed the input file structure to determine the appropriate delimiter (comma, tab, or semicolon), enhancing compatibility with various data sources.

2. **Intelligently map column headers**: Using fuzzy matching for column identification, the system located key fields such as Sequence ID, Nucleotide Sequence, location, host, and release date, regardless of minor variations in header naming.

3. **Filter by geographic location**: Records were processed and sorted by location, with specific filters to extract sequences from our three target states.

4. **Handle data consistency issues**: The system standardized column formats and managed missing values to ensure data integrity.

The output of this process was a set of three structured CSV files (California_data.csv, Texas_data.csv, and New_York_data.csv), each containing the genomic sequences and associated metadata for their respective states.

3.1.2 Sequence Preprocessing

Once the state-specific sequence data was collected, we performed several preprocessing steps to prepare the sequences for alignment:

1. **Sequence cleaning**:

- Converted all sequences to uppercase for standardization
- Removed ambiguous characters (keeping only A, T, C, G)
- Filtered out short sequences (less than 29,000 bp), as these likely represented

incomplete or low-quality samples

2. **FASTA format conversion**:

- Formatted each sequence with a header line starting with "¿" followed by the sequence ID
- Wrapped sequence lines at 80 characters for readability
- Saved each state's cleaned sequences as a separate FASTA file

This preprocessing yielded high-quality datasets for each state, ready for application of our alignment algorithms. The sequence cleaning steps were particularly important for ensuring reliable alignment results, as ambiguous bases or incomplete sequences could lead to alignment errors and misinterpretation of mutation patterns.

3.2 Implementation of MAFFT Algorithm

Our MAFFT implementation followed the four key steps of the algorithm, with modifications to optimize performance for SARS-CoV-2 sequences:

3.2.1 Numerical Encoding and FFT Application

We implemented the Fast Fourier Transform approach to calculate sequence similarities efficiently:

1. **Sequence encoding**: Each nucleotide was assigned a numerical value (A=1, C=2, G=3, T=4) to transform the biological sequences into numerical signals suitable for mathematical operations.

2. **FFT transformation**: We applied the Fast Fourier Transform to these numerical signals, converting them from the time domain to the frequency domain. This transformation captured the pattern of nucleotide changes along the sequence.

3. **Magnitude calculation**: For each transformed sequence, we calculated the magnitude spectrum of the FFT result, which represented the frequency components of the sequence.

4. **Visualization**: We generated plots of the FFT magnitude spectra to visualize and compare the frequency characteristics of different sequences, providing an intuitive representation of sequence similarities and differences.

This approach allowed us to compare sequences based on their frequency characteristics rather than direct sequence comparison, which is particularly efficient for long genomic sequences like SARS-CoV-2.

3.2.2 Distance Matrix Computation

Using the FFT results, we constructed a pairwise distance matrix:

1. **Euclidean distance calculation**: For each pair of sequences, we calculated the Euclidean distance between their FFT magnitude spectra. This distance metric quantified the overall dissimilarity between sequences based on their frequency components.

2. **Matrix construction**: We assembled these distances into a symmetric matrix where each cell (i,j) contained the distance between sequences i and j.

3. **Distance interpretation**: Smaller distances indicated greater sequence similarity, while larger distances suggested more divergent sequences.

This distance matrix served as the foundation for our hierarchical clustering and guide tree construction.

3.2.3 Guide Tree Construction

Based on the distance matrix, we constructed a guide tree to direct the progressive alignment process:

1. **Hierarchical clustering**: We applied the UPGMA (Unweighted Pair Group Method with Arithmetic Mean) algorithm to cluster sequences based on their pairwise distances.

2. **Tree building**: Starting with each sequence as its own cluster, we iteratively merged the closest pairs of clusters, recalculating distances between the new cluster and all other clusters at each step.

3. **Node creation**: Each merging operation created an internal node in the guide tree, representing the ancestral relationship between the merged clusters.

The resulting guide tree represented a hypothesis about the evolutionary relationships between the sequences and determined the order in which sequences would be aligned in the next step.

3.2.4 Progressive Alignment

We implemented a progressive alignment approach following the guide tree topology:

1. **Pairwise alignment**: Starting with the most closely related sequences (as determined by the guide tree), we performed pairwise alignments using a dynamic programming approach similar to the Needleman-Wunsch algorithm.

2. **Profile building**: As sequences were aligned, we created profiles that represented the aligned groups, preserving the gap information.

3. **Profile-profile alignment**: We aligned profiles to each other, following the order dictated by the guide tree, with each profile representing a previously aligned group of sequences.

4. **Gap handling**: We applied appropriate gap penalties for opening and extending gaps, with special consideration for terminal gaps.

This progressive approach ensured that the most similar sequences were aligned first, reducing the likelihood of misalignments in highly variable regions.

3.2.5 Iterative Refinement

To improve the initial alignment, we implemented an iterative refinement procedure:

1. **Sequence removal**: We systematically removed one sequence at a time from the current alignment.

2. **Realignment**: Each removed sequence was realigned to the profile of the remaining sequences.

3. **Evaluation**: We calculated the alignment score before and after realignment to determine if the changes improved the overall alignment.

4. **Iteration**: This process was repeated for all sequences and continued for multiple rounds until no further improvement was observed or a maximum number of iterations was reached.

This refinement step was crucial for optimizing the alignment, especially in regions

where the progressive alignment may have produced suboptimal results.

3.3 Implementation of PRANK Algorithm

Our PRANK implementation incorporated the phylogeny-aware approach with special emphasis on distinguishing between insertions and deletions:

3.3.1 Pairwise Distance Calculation

We calculated evolutionary distances between sequences using a normalized Hamming distance approach:

1. **Hamming distance calculation**: For each pair of sequences, we counted the number of positions where the nucleotides differed.

2. **Normalization**: We divided the count of mismatches by the length of the shorter sequence to obtain a normalized distance value between 0 and 1.

3. **Symmetric matrix construction**: We assembled these normalized distances into a symmetric matrix, ensuring that the distance from sequence A to B was the same as from B to A.

This distance calculation provided a measure of evolutionary divergence between sequences, which formed the basis for our phylogenetic guide tree.

3.3.2 UPGMA Tree Construction

We implemented the UPGMA algorithm to construct a phylogenetic guide tree:

1. **Initialization**: Each sequence started as its own cluster, and all pairwise distances were organized in a priority queue (min-heap) based on their distance values.

2. **Iterative clustering**: At each step, we extracted the pair of clusters with the smallest distance and merged them into a new cluster.

3. **Distance recalculation**: After each merger, we calculated the distance between the new cluster and all other clusters as the average of the distances from each member of the new cluster to each member of the other cluster.

4. **Tree structure maintenance**: We maintained a tree structure that recorded the merging history, with each internal node representing a merger of two clusters.

The resulting guide tree provided a hypothesis about the evolutionary relationships between sequences, which guided the subsequent alignment process.

3.3.3 Pairwise Alignment with Indel Distinction

A key innovation in our PRANK implementation was the distinction between insertions and deletions during alignment:

1. **Dynamic programming approach**: We implemented a variant of the Needleman-Wunsch algorithm with separate gap opening and extension penalties.

2. **Traceback matrix**: In addition to the scoring matrix, we maintained a traceback matrix that recorded the origin of each cell's score (diagonal for match/mismatch, up for deletion, left for insertion).

3. **Gap type annotation**: During the alignment process, we annotated gaps as either insertions or deletions based on the traceback path.

4. **Alignment reconstruction**: We reconstructed the aligned sequences by following the traceback path from the bottom-right cell to the top-left cell, inserting gaps

as needed.

This approach allowed us to maintain information about the nature of gaps (insertions vs. deletions), which is crucial for accurate phylogenetic inference.

3.3.4 Progressive Phylogeny-Aware Alignment

We implemented a progressive alignment strategy that incorporated phylogenetic information:

1. **Profile representation**: We represented partially aligned groups of sequences as profiles, preserving both the sequence information and the gap annotations.

2. **Ancestral sequence reconstruction**: At each internal node of the guide tree, we attempted to reconstruct the ancestral sequence based on the descendant sequences.

3. **Profile alignment**: We aligned profiles to each other, following the guide tree topology from the leaves to the root.

4. **Gap consistency**: We maintained consistency in gap treatments, ensuring that gaps annotated as insertions in one alignment were treated appropriately when that profile was aligned with another profile.

This phylogeny-aware approach resulted in alignments that better reflected the evolutionary history of the sequences, with more accurate placement of gaps.

3.4 Global Alignment Implementation

We implemented the Needleman-Wunsch algorithm for global pairwise alignment, which served as a component of both our MAFFT and PRANK implementations and was also

used for direct pairwise comparisons:

3.4.1 Initialization

We set up the scoring matrix and initialized the first row and column:

1. **Matrix creation**: We created a scoring matrix with dimensions (m+1) × (n+1), where m and n are the lengths of the two sequences being aligned.

2. **First row and column initialization**: We initialized the first row and column with cumulative gap penalties, reflecting the cost of aligning one sequence entirely with gaps.

3. **Scoring parameters**: We used a scoring scheme with +2 for matches, -1 for mismatches, and -2 for gaps, which is appropriate for nucleotide sequence alignment.

3.4.2 Matrix Filling

We filled the scoring matrix using dynamic programming:

1. **Cell-by-cell calculation**: Starting from the upper-left corner, we calculated the score for each cell (i,j) as the maximum of three possibilities:

- Diagonal move: Score from cell (i-1, j-1) plus the match/mismatch score for aligning the characters at positions i and j
- Horizontal move: Score from cell (i, j-1) plus the gap penalty (gap in the first sequence)
- Vertical move: Score from cell (i-1, j) plus the gap penalty (gap in the second sequence)

2. **Score propagation**: Each cell's score depended on previously calculated scores, propagating the optimal alignment choices through the matrix.

3.4.3 Traceback

We reconstructed the optimal alignment by tracing back through the scoring matrix:

1. **Starting point**: We began at the bottom-right cell, which contained the score for the optimal global alignment.

2. **Path determination**: At each step, we determined which of the three possible moves (diagonal, horizontal, or vertical) led to the current cell's score.

3. **Alignment construction**: Based on the traceback path, we constructed the alignment:

- Diagonal move: Align the corresponding characters from both sequences
- Horizontal move: Insert a gap in the first sequence
- Vertical move: Insert a gap in the second sequence

4. **Leading gap handling**: If the traceback ended with remaining positions in either sequence, we inserted leading gaps in the other sequence to complete the alignment.

3.4.4 Identity Calculation

We calculated the sequence identity to quantify the similarity between the aligned sequences:

1. **Match counting**: We counted the number of positions where the two sequences had identical characters.

2. **Percentage calculation**: We divided the number of matches by the total length of the alignment (including gaps) and multiplied by 100 to obtain a percentage identity.

This identity percentage provided a measure of overall sequence similarity, which was useful for comparing sequences from different regions or time periods.

3.5 Phylogenetic Analysis

We implemented a comprehensive phylogenetic analysis to understand the evolutionary relationships between SARS-CoV-2 sequences from different regions:

3.5.1 Multiple Sequence Alignment Processing

We prepared the aligned sequences for phylogenetic analysis:

1. **Alignment format conversion**: We converted our MSA results to a format compatible with phylogenetic analysis tools.

2. **Sequence padding**: We ensured all sequences had the same length by padding shorter sequences with gaps.

3. **Inclusion of reference strain**: We included the Wuhan reference strain (EPI_ISL_402124) in our analysis to serve as an outgroup and reference point.

3.5.2 Distance Calculation

We calculated evolutionary distances between all pairs of sequences:

1. **Distance metric selection**: We used the "identity" distance metric, which calculates distances based on the proportion of non-identical sites between sequences.

2. **Distance matrix construction**: We constructed a symmetric distance matrix containing the pairwise distances between all sequences in the alignment.

3. **Handling of gaps**: We implemented appropriate strategies for dealing with gaps in the alignment during distance calculation.

3.5.3 Tree Construction

We constructed phylogenetic trees using the Neighbor-Joining method:

1. **Neighbor-Joining algorithm**: Starting with a star-like tree, we iteratively joined the closest pairs of nodes, recalculating distances at each step.

2. **Branch length optimization**: We calculated branch lengths to reflect the evolutionary distances between sequences.

3. **Tree topology refinement**: We applied optimizations to improve the tree topology and ensure it accurately represented the evolutionary relationships.

3.5.4 Tree Visualization and Analysis

We implemented methods for visualizing and analyzing the resulting phylogenetic trees:

1. **Tree rendering**: We created graphical representations of the trees with appropriate labeling and scaling.

2. **Branch length interpretation**: We analyzed branch lengths to understand the degree of evolutionary change between sequences.

3. **Clade identification**: We identified major clades (groups of related sequences) within the tree.

4. **Reference strain comparison**: We calculated the evolutionary distance of each regional sequence from the Wuhan reference strain to quantify divergence.

This phylogenetic analysis allowed us to visualize the evolutionary relationships between SARS-CoV-2 sequences from different regions and assess how they had diverged from the original Wuhan strain.

3.6 Consensus Sequence Comparison

We implemented methods to generate and compare consensus sequences from different regions:

3.6.1 Consensus Sequence Generation

We created consensus sequences that represented the dominant viral strain in each region:

1. **Position-by-position analysis**: For each position in the aligned sequences, we determined the most frequent nucleotide.

2. **Majority rule application**: We applied a majority rule, where a nucleotide was included in the consensus if it appeared in more than 50% of sequences at that position.

3. **Gap handling**: If no nucleotide appeared in more than 50% of sequences at a position, we inserted a gap in the consensus sequence.

The resulting consensus sequences represented the "typical" viral genome for each region, facilitating comparisons between regions.

3.6.2 Fast Identity Comparison

We developed an efficient method for comparing consensus sequences:

1. **Gap exclusion**: We excluded positions where either sequence had a gap, focusing only on positions where both sequences had nucleotides.

2. **Match counting**: We counted positions where both sequences had the same nucleotide.

3. **Identity calculation**: We calculated the percentage of matching positions out of all valid comparison positions.

This approach provided a more meaningful measure of sequence similarity than methods that include gaps in the comparison.

3.6.3 Identity Matrix Construction

We compiled the pairwise identity percentages into a comprehensive matrix:

1. **All-vs-all comparison**: We calculated identity percentages for all pairs of consensus sequences.

2. **Matrix assembly**: We organized these percentages into a symmetric matrix with regions as both rows and columns.

3. **Visual presentation**: We formatted the matrix for clear visualization and interpretation.

This identity matrix provided a concise summary of the relationships between viral strains from different regions, highlighting patterns of similarity and divergence.

Chapter 4

Results and Discussion

4.1 Sequence Alignment Results

Our implementation of MAFFT and PRANK algorithms successfully aligned SARS-CoV-2 sequences from California, Texas, and New York. The alignment results revealed different patterns of sequence conservation and variation across the three states.

The MAFFT algorithm, with its FFT-based approach, produced alignments that highlighted conserved regions across the viral genome. These conserved regions likely correspond to functionally important parts of the virus that are under selective pressure to maintain their sequence. Conversely, the algorithm also identified variable regions that showed differences between sequences, which could represent adaptations or neutral mutations.

The PRANK algorithm, with its phylogeny-aware approach, produced alignments that were particularly sensitive to the evolutionary history of the sequences. By distinguishing between insertions and deletions, PRANK provided a more nuanced view of the sequence changes, potentially offering insights into the mechanisms of viral evolution.

A notable observation was that both algorithms identified the spike protein region as having significant variation across sequences, consistent with previous studies showing that this region is subject to strong selective pressure due to its role in host cell entry and immune recognition.

4.2 Consensus Sequence Comparison

The comparison of consensus sequences using the fast identity method revealed interesting patterns:

- Using MAFFT:
  - California and Texas showed high sequence similarity (90.08%)
  - New York showed lower similarity to both California (31%) and Texas (27%)
- Using PRANK:
  - All three states showed lower similarity (26-28%)
  - The lowest similarity was observed between California and Texas (26.63%)

These differences in results between MAFFT and PRANK highlight the impact of algorithmic choices on sequence analysis outcomes. The higher similarity between California and Texas in the MAFFT analysis suggests that these states may have experienced similar viral evolution trajectories, while New York's virus population may have evolved differently.

The discrepancy between MAFFT and PRANK results can be attributed to their different approaches to handling gaps and evolutionary events. MAFFT's FFT-based method may emphasize pattern similarities that are not strictly related to evolutionary history, while PRANK's phylogeny-aware approach may be more sensitive to the specific evolutionary paths taken by the sequences.

These findings underscore the importance of using multiple alignment methods when analyzing viral sequences, as different algorithms may reveal different aspects of sequence relationships.

4.3 Phylogenetic Analysis

The phylogenetic trees constructed using the Neighbor-Joining method provided visual representation of the evolutionary relationships between sequences from different states. The trees also included the Wuhan reference strain, allowing us to assess how far each regional variant had diverged from the original virus.

Several key observations emerged from our phylogenetic analysis:

- Distinct clustering patterns were observed for sequences from different states, suggesting region-specific evolutionary trajectories.
- The California and Texas sequences often formed closely related clades, consistent with the high similarity observed in the consensus sequence comparison.
- New York sequences showed greater evolutionary distance from both the Wuhan reference and the sequences from other states, suggesting potentially different

selection pressures or introduction events.

- Within each state, we observed sub-clades representing different viral lineages, indicating ongoing diversification within regions.

The analysis of drift from the Wuhan reference revealed that New York sequences had, on average, greater evolutionary distance from the reference strain compared to California and Texas sequences. This observation is consistent with the consensus sequence comparison results, further suggesting distinct evolutionary paths for the viral populations in these regions.

These phylogenetic findings provide valuable context for understanding the spread and evolution of SARS-CoV-2 across different U.S. states. The greater divergence observed in New York could be related to earlier introduction events, different host population characteristics, or specific public health measures implemented in the region.

4.4 Regional Variation and Public Health Implications

The observed differences in SARS-CoV-2 sequences across states have important implications for public health:

- Different regions may require targeted surveillance for region-specific variants. Our findings suggest that New York, in particular, may harbor viral variants that have diverged significantly from those in other regions, necessitating specific monitoring efforts.

- Vaccine effectiveness may vary by region if significant mutations affect antigenic sites. The greater sequence divergence observed in New York raises questions about potential impacts on vaccine efficacy in this region, though further analysis of specific mutation sites would be needed to assess this risk.

- Treatment strategies may need to consider regional viral characteristics. As antivirals and other therapeutics are developed, their effectiveness against regionally divergent strains should be evaluated.

- Transmission dynamics may differ between regions due to viral adaptations. The distinct evolutionary patterns observed could reflect adaptations to regional host populations or environmental conditions, potentially affecting transmission rates or severity.

Our analysis identified New York (based on MAFFT results) or California (based on PRANK results) as potentially more prone to harboring divergent strains. This information can guide public health authorities in allocating resources for genomic surveillance and response strategies.

The discrepancy between the MAFFT and PRANK results underscores the importance of using multiple analytical approaches when making public health decisions based on genomic data. By considering results from different alignment methods, authorities can develop more robust and comprehensive response strategies.

Chapter 5

Conclusion

The development and implementation of MAFFT and PRANK algorithms for SARS-CoV-2 sequence analysis have provided valuable insights into regional patterns of viral evolution. Our key findings include:

- Successful implementation of MAFFT and PRANK algorithms from scratch for mutation detection
- Identification of distinct evolutionary patterns in California, Texas, and New York SARS-CoV-2 sequences
- High similarity between California and Texas sequences (90.08%) using MAFFT, suggesting similar evolutionary trajectories
- Lower similarity involving New York sequences (27-31%), indicating greater divergence in this region
- Different results from PRANK implementation, highlighting the importance of algorithm selection in sequence analysis

- Construction of phylogenetic trees to visualize evolutionary relationships and drift from the Wuhan reference

The methodologies developed in this project demonstrate the value of implementing multiple sequence alignment algorithms from scratch, rather than relying solely on existing tools. This approach provided greater transparency, flexibility, and customization for SARS-CoV-2-specific analysis, allowing us to tailor the algorithms to the characteristics of viral genomic sequences.

The comparison between MAFFT and PRANK results revealed that algorithm choice can significantly impact the interpretation of genomic data. MAFFT's FFT-based approach emphasized pattern similarities and identified high similarity between California and Texas, while PRANK's phylogeny-aware approach suggested more uniform divergence across all states. This discrepancy highlights the importance of using multiple analytical methods when studying viral evolution.

Our phylogenetic analysis provided a visual representation of the evolutionary relationships between sequences from different regions and their relationship to the Wuhan reference strain. This analysis supported our sequence comparison findings and offered additional insights into the patterns of viral diversification across regions.

The methodologies developed in this project can be extended to other viral genomic analyses and may aid in epidemiological research and public health response strategies. Future work could include:

- Improving alignment accuracy using additional phylogenetic models

- Automating the process for faster and large-scale mutation analysis
- Extending the study to include more states and international datasets
- Incorporating temporal data to track mutations over time
- Correlating genomic changes with clinical outcomes and epidemiological patterns

This research contributes to our understanding of SARS-CoV-2 genomic variations and demonstrates the effectiveness of computational approaches in tracking viral evolution. The regional patterns of viral diversity identified in this study can inform public health surveillance efforts and guide the development of region-specific response strategies.

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List of Publications based on this research work

1. Group A9, *Comparative Analysis of MAFFT and PRANK Algorithms for SARS-CoV-2 Genomic Sequence Alignment*, International Conference on Bioinformatics and Computational Biology, 2025 (Planned)

2. Group A9, *Regional Patterns of SARS-CoV-2 Evolution: Insights from Multiple Sequence Alignment*, Journal of Viral Genomics, 2025 (In preparation)